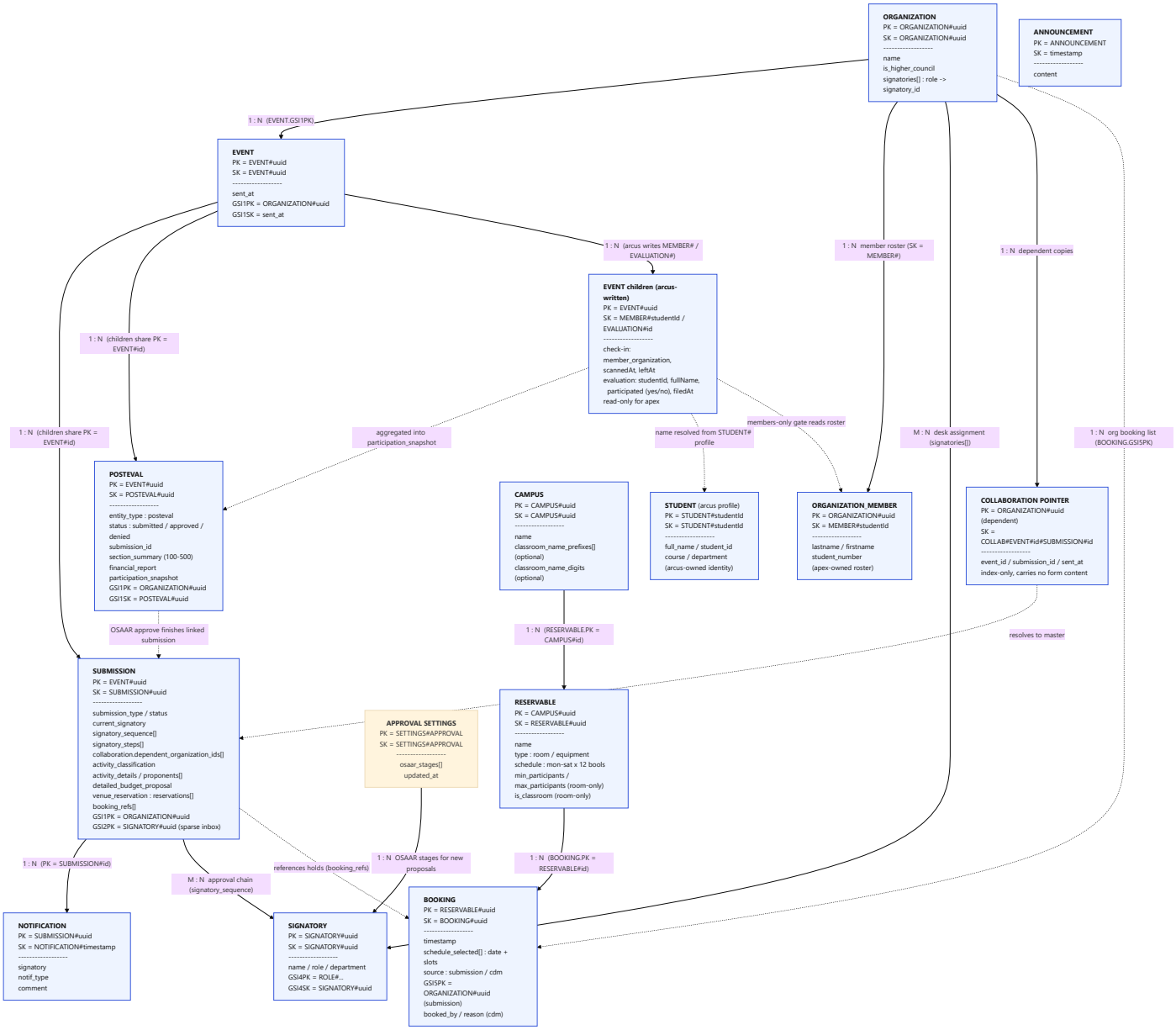
mapua-apex Data Model

DynamoDB single-table data model — conceptual ERD, object dictionary, and access patterns.

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Conceptual ERD



Data Objects & Properties

All objects live in a **single DynamoDB table**. Each item is identified by a partition key (`PK`) and a sort key (`SK`); several item types share one partition so related records are read together in a single query. There are **no foreign keys and no cascade**: related IDs are *copied* onto items and resolved at write time, so deleting one item never rewrites another.

ORGANIZATION

The student organization. Also the desk directory: it records which signatory person occupies each approval role for this org.

Property	Type	Notes
`PK`	string	`ORGANIZATION#uuid`
`SK`	string	`ORGANIZATION#uuid` (same as PK)
`name`	string	Display name
`is_higher_council`	boolean	Default `false` . When true, the Dean step is skipped in routing
`signatories`	list	Desk directory, one person per role: `{ role, signatory_id }`
`signatories[].role`	string	One of `adviser / dean / osaar / cdm / admin`
`signatories[].signatory_id`	string	Plain copy of a SIGNATORY id

Allowed desk roles are stored in the order `adviser → dean → osaar → cdm → admin` regardless of write order. `admin` may hold a desk but is **not** part of the approval sequence. `cdm` is campus-wide: routing uses the single SIGNATORY item whose role is `cdm` (`GSI4PK` `ROLE#CDM`), not a per-org desk. OSAAR stages are the ordered list on `SETTINGS#APPROVAL` (see below). A per-org `osaar` desk is not used to pick the hop.

ORGANIZATION_MEMBER

The organization's member roster, **owned and managed by mapua-apex** (not by the arcus apps). One item per member, sharing the org's partition so the whole roster is read in a single query. Managed through `/api/v1/students/members` and read by the arcus-attendance "Allow members only" scanner toggle to decide whether a scanned student may check in.

Property	Type	Notes
`PK`	string	`ORGANIZATION#uuid` (parent org partition)

Property	Type	Notes
`SK`	string	`MEMBER#studentId`
`lastname`	string	
`firstname`	string	
`student_number`	string	Bare student number (the `SK` without the `MEMBER#` prefix)

Created with an `attribute_not_exists(SK)` guard, so a student number is unique within the org (**409** on duplicate). Bulk CSV import writes in 25-item `BatchWriteItem` chunks and reports duplicates as skipped (`failed[]`) rather than failing the batch (max 500 rows/request). This is the authoritative membership list. It is **distinct** from the arcus STUDENT profile (arcus-owned identity, below) and **replaces** the deprecated arcus-written `ORGANIZATION` + `MEMBER` join rows (same `PK`/`SK` shape but carrying `organization_id` + `student_id` and no names), which `scripts/migrate_member_to_student.py` removes.

EVENT

A filing/event that owns one or more submissions. Its partition also holds the event's SUBMISSION children.

Property	Type	Notes
`PK`	string	`EVENT#uuid`
`SK`	string	`EVENT#uuid` (same as PK)
`sent_at`	timestamp	Creation time (UTC)
`GSI1PK`	string	`ORGANIZATION#uuid` — owning org (also identifies the *proponent* org for collaborations)
`GSI1SK`	timestamp	`sent_at` — for newest-first org event lists

SUBMISSION

A proposal paper (e.g. SAAF) filed under an event. Carries the full form body plus routing state and index keys. This is the single source of truth for a submission, including any collaboration.

Identity, routing & indexes

Property	Type	Notes
`PK`	string	`EVENT#uuid` (parent event partition)
`SK`	string	`SUBMISSION#uuid`
`submission_type`	string	e.g. `saaf`

Property	Type	Notes
`sent_at`	timestamp	Submit time (UTC)
`status`	string	`pending / approved / denied / returned / finished`. `finished` is terminal and set only when OSAAR approves the event's POSTEVAL (see below). `cancelled` is **reserved** for the deferred EventBridge auto-cancel — not written by any code path yet
`current_signatory`	string	`SIGNATORY#uuid` holding the paper now
`signatory_sequence`	list	Ordered `SIGNATORY#uuid` routing snapshot, written at submit time
`signatory_steps`	list	Same snapshot with `{ key, role, signatory_id, organization_id }`. Keys: `adviser:ORG#id`, `dean:ORG#id`, `osaar_1` .. `n`, `cdm`. Absent on items sent before stages were stored; those items are not backfilled
`collaboration.dependent_organization_ids`	list	`ORGANIZATION#uuid` dependents; empty/absent = no collaboration
`GSI1PK`	string	`ORGANIZATION#uuid` (proponent org)
`GSI1SK`	string	`SUBMISSION#uuid`
`GSI2PK`	string	`SIGNATORY#uuid` — sparse signatory inbox (present while `pending` / `returned`)
`GSI2SK`	timestamp	Desk queue time

activity_classification

Property	Type	Notes
`activity_type`	string	`co-curricular / extra-curricular`
`total_org_members`	number	
`nature`	string	`major / minor` — set by OSAAR via the classification PATCH, not by the student create body

proponents[] (list of people)

`id`, `position_title`, `first_name`, `middle_name`, `last_name`, `suffix`, `student_number`, `program_and_year`, `date_of_submission`, `department`, `position_of_applicant`, `org_or_course_section`, `contact_number`, `email_address`, `facebook_link`.

activity_details

`title_and_nature`, `description`, `objectives`, `venue`, `date_of_event`, `day_of_event`, `time_of_event`, `expected_participants`, `individual_contribution`, `proposed_budget`. `time_of_event` is 24-hour `"HH:MM – HH:MM"`; saved times fall between 7:00 AM and 9:00 PM, end not earlier than start, and start ≠ end.

institutional_alignment

`mission_statements { competitive, research, solutions }`, `core_values_explanation`, `peo_explanation`, `sdg_explanation`.

detailed_budget_proposal

`items[] { item_no, unit, quantity, price_per_unit, total }`, `grand_total`.

venue_reservation & booking_refs

Property	Type	Notes
`has_reservation`	boolean	Drives whether the CDM desk joins the routing chain
`reservations[]`	list	One entry per picked reservable: `{ reservable_id, campus_id, name, type, selections[], remarks }`
`reservations[].type`	string	`room` | `equipment`
`reservations[].selections[]`	list	`{ date: "YYYY-MM-DD", slots: [int 0–11] }` — 0-based indices into the 07:00–21:00 / 70-min day (12 slots)
`booking_refs[]`	list	Sparse reverse index `{ pk: "RESERVABLE#uuid", sk: "BOOKING#uuid" }`; present only while the paper holds BOOKINGs. Released + removed on deny/return/cancel-resolution, reconciled on re-edit and on a reschedule-resolution

Legacy items stored before this shape may still carry `equipment_requested` / `function_rooms` / `audiovisual_equipment`; reads tolerate both. On submit each selection is validated against the reservable's weekly template and existing approved BOOKINGs, then written as `source:"submission"` holds with

`approval_status: "pending"` — a slot taken in the meantime aborts with **409** and persists nothing. Pending holds do not occupy slots until the CDM desk approves the paper (see BOOKING).

NOTIFICATION

One row in a submission's approval/timeline history. Stored in a partition keyed by the submission id (a separate partition from the event).

Property	Type	Notes
`PK`	string	`SUBMISSION#uuid`
`SK`	string	`NOTIFICATION#timestamp` (identity; UTC second precision)
`signatory`	string	`SIGNATORY#uuid` author (snapshot)
`notif_type`	string	`approved / fully approved / denied / returned / event scheduled / reservation change requested / reservation cancelled / reservation rescheduled / reservation change denied` — the four `reservation ...` types are system-only (written by RequestReservationChange / ResolveReservationChange, never client-postable)
`comment`	string	Mandatory when `denied` or `returned`

Approve / return / deny insert a NOTIFICATION row as a side effect. A PUT replaces `notif_type` / `comment` in place and does **not** change the SK.

SIGNATORY

A person (the signatory identity). Has **no** `organization_id`; desk assignment lives on `ORGANIZATION.signatories`.

Property	Type	Notes
`PK`	string	`SIGNATORY#uuid`
`SK`	string	`SIGNATORY#uuid` (same as PK)
`name`	string	
`role`	string	`adviser / cdm / dean / osaar / admin`

Property	Type	Notes
`department`	string	Deans only (e.g. `SOIT`)
`GSI4PK`	string	`ROLE#ADVISER` / `ROLE#CDM` / `ROLE#DEAN` / `ROLE#DEAN#{DEPARTMENT}` / `ROLE#OSAAR` / `ROLE#ADMIN`
`GSI4SK`	string	`SIGNATORY#uuid`

Other items copy the signatory id rather than join back to SIGNATORY: `ORGANIZATION.signatories[].signatory_id`, `SUBMISSION.current_signatory` / `GSI2PK`, and `NOTIFICATION.signatory`. Renaming keeps the same uuid; deleting a SIGNATORY does not rewrite those copies. CDM stays one person. Several OSAAR people are allowed; new proposals take their order from APPROVAL SETTINGS.

APPROVAL SETTINGS

One item. Its `osaar_stages` list is how many OSAAR approvals a **new** proposal needs, in that order. Proposals already sent keep the stages stored on their own `signatory_steps`.

Property	Type	Notes
`PK`	string	`SETTINGS#APPROVAL`
`SK`	string	`SETTINGS#APPROVAL`
`osaar_stages`	list	`{ key, signatory_id }`. Keys are `osaar_1` .. `n`, rewritten on each save. Each `signatory_id` is `SIGNATORY#uuid` with role `osaar`, all distinct
`updated_at`	timestamp	Last save

When the item is missing, routing uses the single `ROLE#OSAAR` signatory. Zero OSAAR signatories, or more than one with nothing saved, rejects a new proposal.

COLLABORATION POINTER

An index-only item, one per dependent organization, that makes the proponent's submission discoverable in a dependent's dashboard. It carries **no** form content — the master SUBMISSION stays the single source of truth.

Property	Type	Notes
`PK`	string	`ORGANIZATION#uuid` — the **dependent** org (not the proponent)

Property	Type	Notes
`SK`	string	`COLLAB#EVENT#uuid#SUBMISSION#uuid`
`event_id`	string	`EVENT#uuid`
`submission_id`	string	`SUBMISSION#uuid`
`sent_at`	timestamp	

Queried from the dependent's partition (`SK` begins `COLLAB#`); no GSI is needed because DynamoDB cannot index the N values of an array. A dependent may read the shared submission and its NOTIFICATION timeline but cannot edit or resubmit — only the proponent may.

POSTEVAL

A post-evaluation filed by the org submitter after an approved event is held, within the 3-day evaluation window. It lives in the **event's** partition (alongside its SUBMISSION) and is discovered per-org through GSI1. OSAAR approval is the **only** path that transitions the linked submission `approved → finished`.

Property	Type	Notes
`PK`	string	`EVENT#uuid` (parent event partition)
`SK`	string	`POSTEVAL#uuid`
`entity_type`	string	`"posteval"`
`organization_id`	string	Bare org uuid
`event_id`	string	Bare event uuid
`submission_id`	string	Bare uuid of the linked approved paper
`status`	string	`submitted / approved / denied`
`submitted_at`	timestamp	
`approved_at`	timestamp	Sparse — stamped on OSAAR approve
`denied_at`	timestamp	Sparse — stamped on OSAAR deny
`event_title`	string	Snapshot of `activity_details.title_and_nature`
`section_summary`	string	100–500 chars — how the event went
`financial_report`	object	`{ budget_items[], added_items[], initial_grand_total, actual_grand_total }` ; every amount capped at 5 digits (0–99999)

Property	Type	Notes
`participation_snapshot`	object	`{ attendance_count, attendance[], evaluation_count, evaluations[] }` — frozen at write time
`GSI1PK`	string	`ORGANIZATION#uuid`
`GSI1SK`	string	`POSTEVAL#uuid` (mirrors SUBMISSION for org discovery)

`financial_report.budget_items[]` are `{ item, unit, quantity, budgeted_total, actual_amount }`; `added_items[]` are `{ item, unit, quantity, price_per_unit, actual_amount }`. `participation_snapshot.attendance[]` are `{ student_id, full_name, department, member_organization, scanned_at, left_at }` and `evaluations[]` are `{ evaluation_id, student_id, full_name, participated, filed_at }`.

Write gates (`WritePostEval`): the linked SUBMISSION must be `approved` / `finished` (**422** otherwise), `EventWindow::EVALUATION` must still be open (**422** — "window is closed"), and no `submitted` / `approved` POSTEVAL may already exist for the event (**409**). The snapshot is aggregated from the event's `MEMBER#` / `EVALUATION#` children (below) at write time and stored with an `attribute_not_exists(PK)` guard on the SK.

OSAAR actions: `ApprovePostEval` sets `status = approved` + `approved_at`, then moves the linked SUBMISSION `approved → finished` (+ `finished_at`) — both idempotent, and the window is **not** re-checked (approval may follow the window closing). `DenyPostEval` sets `status = denied` + `denied_at` and leaves the submission `approved`. Approving a denied post-eval (or denying an approved one) aborts **422**.

EVENT-partition children (written by the arcus companion apps)

The `arcus-attendance` and `arcus-evaluation` microservices share the **same** physical table and write into the apex `EVENT#uuid` partition, keyed by the apex `event_id`. apex never writes these — it only reads them (via `AggregateEventParticipants`) to build a POSTEVAL's `participation_snapshot`. Attendance uses the event-day window. Participant evaluations require the signed-in student's check-in and use the evaluation window (event day through three days afterward by default, in Asia/Manila).

Object	PK	SK	Attributes
Attendance check-in	`EVENT#uuid`	`MEMBER#{studentId}`	`member_organization`, `scannedAt`, `leftAt` (sparse)
Participant evaluation	`EVENT#uuid`	`EVALUATION#{evaluationId}`	`evaluationId`, `studentId`, `fullName`, `participated` (`"yes"` / `"no"`), `filedAt`, `eventId`

Arcus Evaluation also writes full response copies at `PK = SK = EVALUATION#{id}` and `PK = EVALUATION`, `SK = {filedAt}#{id}`. These retain the identity/event metadata plus `schemaVersion: 2`, `eventTitle`, all applicable `ratings` (a numeric map for 23 criteria), `discoverySources` (boolean map), `comments`, and `othersSpecify`. Both full copies and the slim event child are committed in one DynamoDB transaction. No new S3 object or bucket/key attribute is written; the browser fills the original FM-SA-05-01 template for preview and export. Legacy metadata without full responses remains visible but cannot regenerate a PDF. No legacy object migration or deletion is performed.

The Arcus officer panel reads live participants and the active POSTEVAL, falling back to the latest denied filing by `submitted_at`. Its `submission_id` resolves the associated paper; orphaned filings are ignored. These reads do not change review status or replace the frozen `participation_snapshot`.

The scanner also keeps a student **profile** item at `PK = SK = STUDENT#{studentId}` holding `student_id`, `full_name`, `course`, `department` (renamed from the former `MEMBER#` profile; the EVENT check-in `SK` above stays `MEMBER#{studentId}` because apex splits an event partition's children by `SK` prefix). apex resolves each check-in's name by reading the `STUDENT#{studentId}` profile. apex reads these camelCase attributes (with snake_case fallbacks) and normalizes `participated` to a boolean in the snapshot.

The "Allow members only" gate reads the apex-owned **ORGANIZATION_MEMBER** roster (`PK = ORGANIZATION#uuid`, `SK = MEMBER#{studentId}`), **not** this profile. arcus registration no longer writes any `ORGANIZATION` + `MEMBER` join row; the former ones are deprecated and removed by `scripts/migrate_member_to_student.py`.

ANNOUNCEMENT

A campus-wide announcement. All announcements share one partition; the SK (timestamp) is the identity. Admin Cognito group only.

Property	Type	Notes
`PK`	string	`ANNOUNCEMENT` (constant)
`SK`	string	`timestamp` — identity (`sent_at` in the API); newest-first list
`content`	string	

A PUT replaces `content` in place and does not change the SK.

CAMPUS

A reservable campus (e.g. `Intramuros Campus`, `Makati Campus`), seeded by osaar on the `/campus` page. The SAAF reservation step and the `activity_details.venue` dropdown read the list.

Property	Type	Notes
`PK`	string	`CAMPUS#uuid`

Property	Type	Notes
`SK`	string	`CAMPUS#uuid` (same as PK)
`name`	string	Display name
`classroom_name_prefixes`	list	**Optional.** Allowed classroom-name prefix tokens (1–10 letters each, up to 30; normalized uppercase + deduped). Stored only when paired with `classroom_name_digits`
`classroom_name_digits`	number	**Optional.** Required digit count (1–4). With the prefix list it defines how this campus's classroom rooms are named (e.g. `MPO` + 3 → `MPO123`)

List scans `PK` begins `CAMPUS#` AND `PK = SK`. Delete is guarded: **409** if any RESERVABLE still lives under `PK = CAMPUS#id`. Rename in place (PUT) to keep the same uuid so RESERVABLE children and stored booking `campus_id` snapshots stay valid. The classroom format is authored on the osaar `/campus` Add/Edit page; a campus without it cannot host classroom rooms.

RESERVABLE

A bookable room or equipment item under a campus. Its `SK` is globally unique and doubles as the BOOKING partition key.

Property	Type	Notes
`PK`	string	`CAMPUS#uuid` (parent campus)
`SK`	string	`RESERVABLE#uuid`
`name`	string	
`type`	string	`room` | `equipment`
`schedule`	object	Recurring weekly template: `monday` .. `saturday`, each exactly 12 booleans (one per 07:00–21:00 / 70-min slot; `true` = available). Sunday is never reservable; an absent day is all-unavailable
`min_participants`	number	**Room-only, optional.** Lower bound on the expected headcount a room can hold. Omitted/unset = no lower limit; never stored for `equipment`
`max_participants`	number	**Room-only, optional.** Upper bound on the expected headcount a room can hold. Omitted/unset = no upper limit; never stored for `equipment`

Property	Type	Notes
`is_classroom`	boolean	**Room-only, optional.** True marks the room as a classroom whose `name` must match the owning campus's classroom format. Stored only when `type` is `room` and the flag is true; read back as false otherwise

Added/edited by cdm on the `/reservables` page (CSV for name+type, a grid for the schedule; participant bounds are room-only inputs, disabled and dropped for equipment). The `is_classroom` toggle is room-only and shown-but-disabled until the selected campus defines a classroom format; when enabled the room `name` is validated against `<one prefix><exactly N digits>` at create/update (**422** on the name if it misses, or **422** on `is_classroom` if the campus has no format). On SAAF submit the activity's `expected_participants` is checked against each picked room's bounds (`BookingRecords::assertAvailable`); an out-of-range headcount aborts with **422** before any BOOKING is written. Delete is guarded: **409** if any BOOKING still occupies `PK = RESERVABLE#id` .

BOOKING

A concrete hold on a reservable's slots for specific dates. One BOOKING per (reservable, submission) or per (reservable, CDM reserve action). Two flavors share the item type, distinguished by `source` .

Property	Type	Notes
`PK`	string	`RESERVABLE#uuid`
`SK`	string	`BOOKING#uuid`
`timestamp`	timestamp	Create time (UTC)
`schedule_selected`	list	`[{ date: "YYYY-MM-DD", slots: [int 0-11] }]` (normalized)
`source`	string	`submission` (SAAF-created hold) | `cdm` (manual CDM hold)
`campus_id`	string	Bare campus uuid (snapshot)
`reservable_name`	string	Snapshot at write time
`reservable_type`	string	`room` | `equipment` (snapshot)

Submission-sourced (`source: "submission"`) adds GSI5 + traceability, and is released on deny/return/re-edit via the SUBMISSION's `booking_refs` :

Property	Type	Notes
`GSI5PK`	string	`ORGANIZATION#uuid` — owning org (org booking list)
`GSI5SK`	timestamp	`timestamp`

Property	Type	Notes
`event_id` / `submission_id` / `organization_id`	string	Bare uuids for traceability and read-only CDM labelling
`approval_status`	string	`pending` (written on SAAF submit/re-edit) | `approved`. Absent on legacy items, which read as approved
`approved_at`	timestamp	Sparse — stamped when CDM approves the paper
`approved_by`	string	Sparse — `SIGNATORY#uuid` of the approving CDM signatory

Only **approved** holds occupy slots. On the paper's last approval (the CDM desk, which is always last when a venue is reserved) `ApproveSubmission` re-checks every `booking_refs` hold against the approved bookings — a slot taken in the meantime aborts **409** with nothing written, leaving the paper on the CDM desk to return or deny — then flips each hold to `approved`. The CDM review screen reads the same check (without writing) from `GET /signatories/events/{event}/submissions/{submission}/booking-conflicts`. Pending holds are still listed on the CDM Reserve/View page (labelled) and still block reservable deletion. Legacy holds are stamped by `scripts/backfill_booking_approval.py`.

Manual (`source: "cdm"`) carries **no** GSI5 and **no** submission linkage (so it never surfaces in an org's list); it is released only by CDM delete:

Property	Type	Notes
`booked_by`	string	`SIGNATORY#uuid` (the CDM signatory)
`reason`	string	Optional purpose note (omitted when blank)

Both flavors share one availability engine: a slot is free only when the weekly template allows it AND no approved BOOKING (either source; manual holds always count) occupies that date+slot. Conflict detection runs on the base RESERVABLE partition — never GSI5 — so it works before the (operator-provisioned) organization index exists.

A submission-sourced hold is also **edited in place** through a RESERVATION_CHANGE resolution: a same-reservable date/slot change patches `schedule_selected` on the existing `PK` / `SK` (keys, GSI5 linkage and `submission_id` / `event_id` / `organization_id` all survive, and the SUBMISSION's `booking_refs` entry stays valid); a room swap deletes the old BOOKING and recreates a `source:"submission"` hold under the new RESERVABLE partition, swapping the `booking_refs` entry to the new keys in the same submission update. The recreated hold is written `approval_status: "approved"` (the paper is already approved).

RESERVATION_CHANGE

A student's request to cancel or reschedule the holds on an **approved** paper (which the student cannot edit directly), resolved by CDM. Each request freezes the concrete BOOKING records it targets and keeps a full before/after audit trail once resolved.

Property	Type	Notes
`PK`	string	`EVENT#uuid` (parent event partition)
`SK`	string	`RESERVATION_CHANGE#uuid`
`item_type`	string	`reservation_change`
`organization_id` / `submission_id` / `campus_id`	string	Bare uuids, snapshotted at request time
`requested_by`	string	`STUDENT#<sub>` from the JWT
`requested_at`	timestamp	
`request_type`	string	`cancel` | `reschedule`
`reason`	string	Required (≤500)
`suggested_date`	string	**Sparse, reschedule only.** `"YYYY-MM-DD"`, advisory — CDM is not bound by it
`bookings[]`	list	The targeted holds, **frozen** from the paper's live `booking_refs` at request time: `{ pk, sk, reservable_id, reservable_name, reservable_type, schedule_selected[] }`
`status`	string	`pending / resolved / denied`
`GSI6PK`	string	`CAMPUS#uuid` — CDM per-campus queue
`GSI6SK`	string	`status#requested_at` (`pending#…`, `resolved#…`, `denied#…`)

On resolution the item also carries `decision` (`cancel / reschedule / deny`), `resolved_by` (`SIGNATORY#uuid`), `resolved_at`, `comment`, and an `applied` audit object:

Property	Type	Notes
`applied.bookings[]`	list	One entry per targeted hold: `{ pk, sk, action: released | kept | updated | replaced, before: { reservable_id, schedule_selected }, after: {…} | null, new_pk, new_sk }` — `new_pk` / `new_sk` only on a room swap (point at the recreated BOOKING)

Property	Type	Notes
`applied.submission`	object | null	`{ status_before, status_after, date_of_event_before/after, time_of_event_before/after, booking_refs_before, booking_refs_after }`; `null` for a deny

`applied` is an audit trail only — never read back as the source of truth. One pending request per paper. The student side lists the full history with a base-table query on `PK = EVENT#uuid`, `SK` begins `RESERVATION_CHANGE#` (no GSI); the CDM queue reads **GSI6** (operator-provisioned in AWS, mirroring GSI5; the CDM list endpoint fails loudly until it exists). Resolve is guarded: the request must still be `pending` (`finalize` uses a conditional put on `attribute_not_exists(resolved_at)` so two CDM users cannot resolve it twice), the paper must still be `approved`, and every snapshotted hold must still appear in the paper's live `booking_refs` (a stale request aborts **409** with zero writes). The request window closes at the end of `activity_details.date_of_event` in Asia/Manila; CDM may resolve at any time.

mapua-apex — DynamoDB Access Patterns

DynamoDB is modeled around **access patterns**, not normalized relations: every query is designed first, then the partition key (PK), sort key (SK), and global secondary indexes (GSI) are chosen to serve it. All patterns hit a **single table**. There are no joins — related IDs are copied onto items at write time.

Index summary:

Index	Key	On	Purpose
Base table	`PK` / `SK`	all items	Primary access
GSI1	`GSI1PK` / `GSI1SK`	EVENT, SUBMISSION, POSTEVAL	Org event walk (deadlines), student & admin submission lists, org/OSAAR post-eval list
GSI2	`GSI2PK` / `GSI2SK`	SUBMISSION	Sparse signatory inbox (`pending` / `returned`)
GSI4	`GSI4PK` / `GSI4SK`	SIGNATORY	Role directory (`ROLE# ...`)
GSI5	`GSI5PK` / `GSI5SK`	BOOKING	Org-scoped booking list (submission-sourced holds only; operator-provisioned)
GSI6	`GSI6PK` / `GSI6SK`	RESERVATION_CHANGE	CDM per-campus reservation-change queue (`status#requested_at` ; operator-provisioned)

Core access patterns

#	Access pattern	Operation	Key condition	Index
1	Org event walk (deadlines list)	Query	`GSI1PK = ORGANIZATION#id` (first 25 EVENT items, oldest first)	GSI1
2	Event children (its deadlines)	Query	`PK = EVENT#id` , `SK` begins `DEADLINE#`	Base
3	Student's own submissions	Query	`GSI1PK = ORGANIZATION#id` , `GSI1SK` begins `SUBMISSION#` (all papers, paginated)	GSI1

#	Access pattern	Operation	Key condition	Index
4	Collaboration submissions (dependent view)	Query	`PK = ORGANIZATION#<dep>`, `SK` begins `COLLAB#`, then resolve each pointer to its master SUBMISSION	Base
5	Single submission detail	GetItem	`PK = EVENT#id`, `SK = SUBMISSION#id`	Base
6	Submission timeline / audit	Query	`PK = SUBMISSION#id`, `SK` begins `NOTIFICATION#`	Base
7	Signatory inbox (desk queue)	Query	`GSI2PK = SIGNATORY#id`, `GSI2SK` by time (sparse)	GSI2
8	Signatory directory by role	Query	`GSI4PK = ROLE#...`, `GSI4SK` begins `SIGNATORY#`	GSI4
8b	OSAAR approval stages	GetItem / PutItem	`PK = SK = SETTINGS#APPROVAL` (`osaar_stages`). Missing item falls back to the single `ROLE#OSAAR` signatory	Base
9	Current signatory profile (`/signatories/me`)	GetItem	`PK = SIGNATORY#id`, `SK = SIGNATORY#id`	Base
10	Admin: all submissions	Query	per org `GSI1PK = ORGANIZATION#id`, `GSI1SK` begins `SUBMISSION#`	GSI1
11	Student org record (`/students/organization`)	GetItem	`PK = ORGANIZATION#id`, `SK = ORGANIZATION#id`	Base
12	Org directory (`/students/organizations`)	Query	org list → `{ organization_id, name }`	Base

#	Access pattern	Operation	Key condition	Index
13	Announcement list	Query	`PK = ANNOUNCEMENT`, `SK` desc (newest first)	Base
14	Announcement show / edit / delete	GetItem / PutItem / DeleteItem	`PK = ANNOUNCEMENT`, `SK = timestamp`	Base
15	Write submission (+ collab pointers)	PutItem	`PK = EVENT#id`, `SK = SUBMISSION#id`; maintain one COLLAB pointer per dependent	Base
16	Approve / return / deny	UpdateItem	advance `current_signatory` / `GSI2PK` from stored `signatory_sequence`; write NOTIFICATION	Base
17	Admin: delete organization (guarded)	Query → DeleteItem	block (**409**) if `GSI1PK = ORGANIZATION#id` has events/submissions or base `PK = ORGANIZATION#id`, `SK` begins `COLLAB#`; else DeleteItem `PK = SK = ORGANIZATION#id`	GSI1 + Base
18	Admin: delete signatory (guarded)	Scan/Query → DeleteItem	block (**409**) if any `ORGANIZATION.signatories` desk holds the id or `GSI2PK = SIGNATORY#id` (open inbox); else DeleteItem `PK = SK = SIGNATORY#id`	GSI2 + Base
19	Campus list (osaar `/campus`, SAAF venue dropdown)	Scan	`PK` begins `CAMPUS#` AND `PK = SK`	Base
20	Reservables under a campus	Query	`PK = CAMPUS#id`, `SK` begins `RESERVABLE#`	Base

#	Access pattern	Operation	Key condition	Index
21	Reservable availability (template + per-date free/booked slots)	Query	`PK = RESERVABLE#id`, `SK` begins `BOOKING#`; overlay approved bookings (pending SAAF holds skipped) on the weekly template across the window	Base
22	CDM View/Reserve calendar (raw bookings, both sources)	Query	`PK = RESERVABLE#id`, `SK` begins `BOOKING#` (filtered to dates in window)	Base
23	Create booking (submission hold or CDM manual)	Query → PutItem	validate selections vs template + existing approved bookings (**409** on conflict) and, on submission, expected headcount vs each room's `min/max_participants` (**422** out of range), then PutItem `PK = RESERVABLE#id`, `SK = BOOKING#id`	Base
24	Release holds on deny / return / re-edit / cancel-resolution	DeleteItem	delete each `booking_refs[] {pk, sk}` (submission holds)	Base
25	Delete a manual booking (CDM)	GetItem → DeleteItem	**409** if `source` is `submission`; else DeleteItem `PK = RESERVABLE#id`, `SK = BOOKING#id`	Base
26	Admin: delete campus / reservable (guarded)	Query → DeleteItem	block (**409**) if a campus owns any RESERVABLE, or a reservable owns any BOOKING; else DeleteItem	Base

#	Access pattern	Operation	Key condition	Index
27	Create / edit reservable (CDM Add/Edit)	GetItem → PutItem	for a `room` flagged `is_classroom`, GetItem the owning CAMPUS and enforce it defines a classroom format (**422** on `is_classroom` if not) and the `name` matches `<prefix> <N digits>` (**422** on `name`), then PutItem `PK = CAMPUS#id`, `SK = RESERVABLE#id`; room-only `min/max_participants` + `is_classroom` dropped for `equipment`	Base
28	Eligible post-eval events (student step 1)	Query	per approved/finished submission compute the effective date (booking date when reserved, else `date_of_event`), keep those already passed with `EventWindow::EVALUATION` open, minus events that already have a `submitted` / `approved` POSTEVAL	GSI1 + Base
29	Org post-eval list (student dashboard)	Query	`GSI1PK = ORGANIZATION#id`, `GSI1SK` begins `POSTEVAL#`	GSI1
30	OSAAR post-eval queue (cross-org)	Query	per org `GSI1PK = ORGANIZATION#id`, `GSI1SK` begins `POSTEVAL#` (optional `status` filter)	GSI1
31	Single post-eval detail	GetItem	`PK = EVENT#id`, `SK = POSTEVAL#id`	Base

#	Access pattern	Operation	Key condition	Index
32	Aggregate event participants (post-eval Section 3)	Query	`PK = EVENT#id`; split `SK` begins `MEMBER#` (attendance; resolve names via `PK = SK = STUDENT#{studentId}`) vs `EVALUATION#`	Base
33	Write post-eval (gated)	Query → PutItem	enforce submission approved/finished + window open (**422**) + no active POSTEVAL (**409**), freeze the snapshot, PutItem `PK = EVENT#id`, `SK = POSTEVAL#id` (`attribute_not_exists(PK)`)	Base + GSI1
34	Approve / deny post-eval (OSAAR)	UpdateItem	POSTEVAL `submitted → approved` / `denied`; on approve also SUBMISSION `approved → finished` (idempotent)	Base
35	Arcus student: attended evaluatable events	Query → BatchGetItem	List published evaluation-window submissions across orgs, then check `PK = EVENT#id`, `SK = MEMBER#{verifiedStudentId}` in deduplicated batches of 100; retry unprocessed keys, fail on exhaustion	GSI1 + Base
36	Arcus officer: live participants + review status	Query	`PK = EVENT#id`, prefixes `SUBMISSION#` and `POSTEVAL#`; prefer active/latest filing and its linked submission, enforce org scope, then aggregate participants (32)	Base

#	Access pattern	Operation	Key condition	Index
37	Arcus student: file full response	TransactWriteItems	Full detail `PK = SK = EVALUATION#id`; full list `PK = EVALUATION`, `SK = filedAt#id`; slim `EVENT#id/EVALUATION#id` child, all conditional puts	Base
38	Arcus officer: event-scoped export	Query → BatchGetItem	Query `EVENT#id`, `SK` begins `EVALUATION#`, then batch-read full `EVALUATION#id` detail records; filter/sort, return up to 100; PDF generation stays in browser	Base
39	CDM approval of a paper's holds	GetItem → UpdateItem	on the last (CDM) approve, GetItem each `booking_refs[]` hold + its RESERVABLE, re-check against approved bookings (**409**, nothing written, on conflict), then UpdateItem `approval_status = approved`, `approved_at`, `approved_by`; the same check (no writes) backs `GET …/booking-conflicts`	Base

Page / feature → objects accessed

Page / feature	Objects read or written	Patterns
Student dashboard	ORGANIZATION, EVENT, SUBMISSION, ANNOUNCEMENT, COLLAB POINTER, POSTEVAL	1, 2, 3, 4, 11, 13, 29
Submission tracker / detail	SUBMISSION, NOTIFICATION, SIGNATORY (by copied id)	5, 6

Page / feature	Objects read or written	Patterns
New / edit submission (SAAF)	SUBMISSION, COLLAB POINTER, ORGANIZATION (directory), CAMPUS, RESERVABLE, BOOKING (holds + `booking_refs`)	12, 15, 19, 20, 21, 23, 24
Signatory desk (inbox)	SUBMISSION, SIGNATORY	7, 9
Signatory review (approve/return/deny/classify)	SUBMISSION, NOTIFICATION, BOOKING (release holds on deny/return; approve holds on CDM approve), RESERVABLE	6, 16, 24, 39
Admin dashboard (submissions)	ORGANIZATION, SUBMISSION	10
Admin → Organizations	ORGANIZATION, SIGNATORY	11, 12, 17
Admin → Signatories	SIGNATORY, ORGANIZATION (desk guard)	8, 18
osaar → Campus	CAMPUS, RESERVABLE (delete guard)	19, 20, 26
cdm → Reservables (Add/View/Reserve)	CAMPUS, RESERVABLE, BOOKING	20, 21, 22, 23, 25, 26, 27
Student dashboard → request change (dialog)	SUBMISSION, RESERVATION_CHANGE, BOOKING (snapshot), NOTIFICATION	6, 28, 29
cdm → Reservations (resolve cancel / reschedule / deny)	RESERVATION_CHANGE, SUBMISSION, BOOKING, RESERVABLE, CAMPUS, NOTIFICATION	20, 21, 23, 24, 30, 31
About / Announcements	ANNOUNCEMENT	13, 14
Student → Post-evals (entry + wizard)	SUBMISSION, POSTEVAL, EVENT children (MEMBER#/EVALUATION#)	28, 31, 32, 33
Student dashboard → Post-eval panel	POSTEVAL	29
OSAAR → Post-evals (queue + review)	POSTEVAL, SUBMISSION (finish on approve)	30, 31, 32, 34
Arcus student → My events / form	SUBMISSION, attendance child, STUDENT profile, full/slim EVALUATION records	35, 37
Arcus officer → Event workspace / export	SUBMISSION, POSTEVAL, attendance/evaluation children, full EVALUATION records	32, 36, 38